Intro

Financial markets transition through identifiable regimes defined by direction and volatility state, each with distinct risk characteristics and optimal trading behaviour. Financial market regimes refer to clusters of persistent market conditions that shape overall market behavior. These regimes describe whether markets are trending upward (bull) or downward (bear), moving sideways, or experiencing periods of high or low volatility (figure 1). Understanding these patterns is important because different market environments favor different investment strategies. A strategy that performs well in a calm, low-volatility environment may struggle or fail entirely during periods of heightened volatility or market stress. As a result, identifying the current market regime can help investors and quantitative models adjust risk management, position sizing, and strategy selection more effectively. Since regime identification is inherently lagging and volatility dynamics are nonlinear, a static or single-model framework is insufficient for robust forecasting and risk management. This project develops a research grade quantitative system for regime-conditional volatility forecasting and portfolio construction that converts equity option chains into implied volatility surface representations and integrates them with realized return dynamics to generate regime-aware forecasts and tradable portfolio overlays. The main idea behind this project is to treat the implied volatility surface as structured spatial data and encode it using a Vision Transformer. This enables the model to extract nonlinear patterns across moneyness and maturity dimensions. The learned surface representation is fused with a temporal encoder applied to realized returns and marker state variables, allowing the system to jointly model cross-sectional option information and time-series dynamics. The multimodal architecture produces forward-looking risk signals, including realized volatility forecasts, tail-risk probabilities, and latent regime classifications. These signals are mapped into a transparent, rules-based trading overlay that dynamically adjusts portfolio exposure in response to changing market conditions. The final system is constructed to be economically interpretable, statistically robust, and practically implementable using publicly available data, ensuring both academic validity and real-world applicability.

	Quiet	Volatile
Bull	Bull quiet	Bull volatile
Bear	Bear quiet	Bear volatile
Sideways	Sideways quiet	Sideways volatile

Figure 1. Marget regimes defined by direction and volatility state

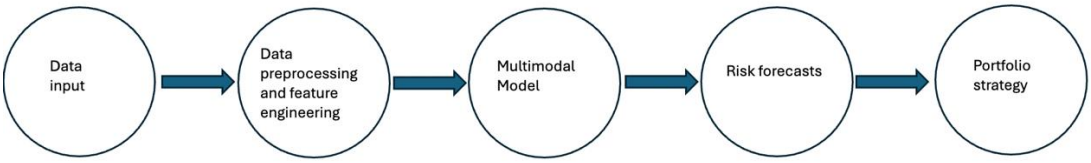

Figure 2. End-to-end pipeline for regime-control volatility forecasting

Background & Related Work

Financial markets do not move through a single stable state. Periods of calm trading, persistent trends, and stress-driven reversals tend to cluster, which is why regime-based thinking has long been central in financial modeling. Regime-switching work, especially the Markov-switching tradition associated with Hamilton, formalized the idea that the statistical behavior of a time series can change abruptly across latent states rather than drift smoothly around one fixed mean. In parallel, volatility modeling research showed that return variance is itself time-varying and clustered, which made constant-variance models inadequate for risk forecasting and portfolio design (Bollerslev, 1986; Hamilton, 1989).

A major branch of this literature focuses on forecasting realized volatility from historical returns. GARCH-type models remain a standard benchmark because they explicitly model conditional variance persistence and capture volatility clustering with relatively low complexity (Bollerslev, 1986). Later work extended this idea with multi-horizon structures. In particular, the HAR-RV model represents realized volatility as an additive cascade across daily, weekly, and monthly components, offering a parsimonious way to reproduce long-memory behavior while preserving strong empirical forecasting performance (Corsi, 2009). These models are useful baselines, but they are backward-looking by construction. Their forecasts are driven mainly by past realized behavior, which can limit responsiveness when market conditions shift quickly or when option markets embed forward-looking information not yet visible in returns alone (Bollerslev, 1986; Corsi, 2009).

This limitation is one reason option-implied information is so important. Implied volatility is extracted from option prices and reflects the market's expectation of future uncertainty under current pricing conditions. At the index level, the VIX is explicitly designed to measure 30-day forward-looking expected equity market volatility using S&P 500 option prices (Cboe Global Indices, 2026). More generally, the implied volatility surface organizes option-implied information across moneyness and time to maturity, making it a richer object than any single at-the-money volatility measure. Prior work has shown that the surface contains information about return distributions, tail shape, skew, and term structure, while also creating practical

challenges because real option chains are sparse, noisy, and not automatically arbitrage-free (Cont & da Fonseca, 2002; Homescu, 2011; Cboe Global Indices, 2026).

Classical work on implied volatility surfaces often tries to reduce this object into a lower-dimensional representation. Cont and da Fonseca (2002) showed that surface movements can be summarized by a small number of dominant factors, which made factor-based and principal-component style approaches attractive. More practitioner-oriented work has also emphasized interpolation, smoothing, and arbitrage constraints when constructing usable surfaces from incomplete market quotes (Homescu, 2011). These approaches are useful for interpretation and calibration, but they can compress away local nonlinear structure. That matters because the geometry of the surface is not merely a nuisance variable. It may contain predictive patterns tied to how risk is priced across strikes and maturities at the same time (Cont & da Fonseca, 2002; Homescu, 2011).

Recent machine learning studies push further by modeling the surface directly instead of relying only on handcrafted summaries. Cao, Chen, and Hull (2019) show that neural networks can capture nonlinear relationships between index returns, moneyness, maturity, and implied volatility movements more effectively than a simpler analytic benchmark. Their results are especially relevant here because they find that the relationship between returns and surface changes depends on the prevailing volatility level. In low-volatility environments, large positive index moves can generate implied volatility responses that differ from the standard leverage-style intuition, which supports the view that volatility dynamics are state-dependent rather than globally stable (Cao et al., 2019). This is closely aligned with the motivation for regime-conditional forecasting.

A more recent strand treats the implied volatility surface as image-like data. Kelly, Kuznetsov, Malamud, and Xu (2023) apply deep learning methods from computer vision to option-implied volatility surfaces and show that these models extract predictive information that is largely uncorrelated with standard option-implied characteristics. They also report that ensembling materially improves out-of-sample performance and argue that simple low-rank rotations of the surface do not fully capture the information content embedded in the full cross-sectional shape (Kelly et al., 2023). This is an important shift in the literature because it suggests that the surface should not only be smoothed or factorized, but can instead be learned as a structured spatial object.

Related evidence also comes from the broader deep volatility forecasting literature. Ge et al. (2023) compare several deep learning models for volatility prediction using multivariate inputs and find that architectures such as the Temporal Fusion Transformer and Temporal Convolutional Networks often outperform naive and classical GARCH-based benchmarks. However, that line of work generally uses time-series and tabular data rather than a full option-

surface representation (Ge et al., 2023). As a result, it supports the case for deep sequential models, but it does not by itself solve the problem of extracting information from cross-sectional option geometry.

Taken together, the literature points to three clear conclusions. First, volatility is regime-dependent and difficult to model well with a single static specification (Bollerslev, 1986; Hamilton, 1989). Second, implied volatility surfaces contain forward-looking information that is richer than historical returns or scalar volatility indicators alone (Cont & da Fonseca, 2002; Homescu, 2011; Cboe Global Indices, 2026). Third, deep learning methods can capture nonlinear patterns in both volatility time series and option-surface data that simpler benchmarks miss (Cao et al., 2019; Kelly et al., 2023; Ge et al., 2023). The gap is that these strands are still only partially integrated. Existing studies often focus on either historical volatility forecasting, surface construction and interpretation, or image-based return prediction from options. There is much less work that jointly uses the full implied volatility surface, realized return dynamics, and market-state variables to produce direct regime-conditional volatility and tail-risk forecasts, then map those forecasts into a transparent portfolio overlay. That is the space this project occupies. By fusing a surface encoder with a temporal returns model and an explicit regime-aware decision layer, the proposed system aims to connect prediction accuracy, market structure, and practical tradability in one framework (Corsi, 2009; Cao et al., 2019; Kelly et al., 2023; Ge et al., 2023).

Methodology

System Architecture:

Data Preprocessing and Feature Engineering:

The project integrates four primary datasets: option chain data, underlying price data, volatility history data, and market state data. Each dataset undergoes dedicated preprocessing and feature engineering before being combined into multimodal inputs for the forecasting model.

The option chain dataset forms the primary input for constructing the implied volatility surfaces used in the model. Option datasets can often contain inconsistent formatting and missing observations, so a standardized filtering pipeline was implemented. First, option type indicators were normalized to a consistent representation. Column naming inconsistencies were also standardized. Quality control filters are applied to remove unreliable contracts, invalid bid-ask relationships, extreme implied volatility values, excessive bid-ask spreads, and deltas outside reasonable bounds. These steps are to help eliminate illiquid or stale quotes that could distort the volatility surface. After the data cleaning, the option data is joined with underlying spot prices to compute additional structural features. Additional filtering ensures the dataset only contains contracts within a specified maturity and moneyness range. Next, the filtered options

are mapped onto a fixed grid of maturity buckets and moneyness bins to construct implied volatility surface tensors. Multiple feature channels are included, such as implied volatility, normalized bid-ask spread, observation masks indicating data availability, and option Greeks.

The underlying price data is used to create time-series features representing return dynamics. Rolling realized volatility estimates are produced across multiple time horizons by computing log returns from adjusted closing prices. Additional features include rolling beta estimates in relation to a broad market index to gauge systematic risk exposure and jump indicators to identify unusual price fluctuations. These features are organized into a fixed-length lookback window, forming a returns tensor that captures recent price behaviour and volatility patterns for each asset.

The volatility history dataset includes contextual information regarding the relative level of behaviour and volatility over time. Implied volatility rank, historical volatility rank, volatility risk premium, short and medium-term volatility momentum measures, and indicators measuring the number of days since recent volatility highs and lows are among the features extracted from this dataset. These features aid in determining if volatility is elevated, declining, or near historical extremes, which offers helpful regime context for forecasting.

Market state data provides macro-level information that may have an impact on the volatility dynamics of certain assets. Key features include the volatility index, market index returns, and optionally the risk-free interest rate. These factors offer insights into systematic risk conditions, the overall market sentiment, and the broader economic environment. This dataset is converted into a condensed feature vector that adds market-wide information to the model in addition to individual option surfaces.

Finally, targets are constructed for supervised learning. Forward realized volatility is computed over a fixed prediction horizon and transformed using a logarithmic scale for stability. A tail-risk classification indicator and a rule-based market-regime label based on trend direction, volatility circumstances, and trend strength indicators are additional targets. The model is trained on reliable, high-quality inputs that capture both cross-sectional option information and temporal market dynamics thanks to this data processing pipeline.

References

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics, 31(3), 307-327. https://doi.org/10.1016/0304-4076(86)90063-1

Cao, J., Chen, J., & Hull, J. (2019). A neural network approach to understanding implied volatility movements. Rotman School of Management Working Paper.

Cboe Global Indices. (2026). Volatility index methodology: Cboe volatility index (Version 6.0). Cboe Global Markets.

Cont, R., & da Fonseca, J. (2002). Dynamics of implied volatility surfaces. Quantitative Finance, 2(1), 45-60. https://doi.org/10.1088/1469-7688/2/1/304

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. Journal of Financial Econometrics, 7(2), 174-196. https://doi.org/10.1093/jjfinec/nbp001

Ge, W., Lalbakhsh, P., Isai, L., Lensky, A., & Suominen, H. (2023). Comparing deep learning models for the task of volatility prediction using multivariate data. arXiv preprint arXiv:2306.12446.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica, 57(2), 357-384. https://doi.org/10.2307/1912559

Homescu, C. (2011). Implied volatility surface: Construction methodologies and characteristics. arXiv:1107.1834.

Kelly, B., Kuznetsov, B., Malamud, S., & Xu, T. A. (2023). Deep learning from implied volatility surfaces. Working paper.